Classical ML vs. Deep Learning for CVD Classification on NHANES 2011–2018 — Experimental Results

All numbers below come from the scripts in this folder (run order `01` → `06`) and are reproducible (seed 42). Tables are in `results/tables/` (CSV + Markdown), figures in `figures/` (300-dpi PNG + vector PDF).

1. What was done (Methodology section material)

Data. Four NHANES cycles: 2011–2012 (G), 2013–2014 (H), 2015–2016 (I), 2017–2018 (J), downloaded from CDC (files DEMO, BPX, BMX, MCQ, TCHOL, HDL, GHB, BIOPRO, SMQ, DIQ; BPQ used only for the supplementary medication check). All variable names were verified to exist with identical coding in all four cycles. 2019–2020 was excluded because collection was interrupted by COVID-19.

Population. Adults ≥ 20 y who completed the mobile examination (RIDSTATR = 2) and had a determinable outcome.

Cycle	Participants	Examined adults ≥20	Outcome determined	CVD cases
2011–2012	9,756	5,319	5,302	550
2013–2014	10,175	5,588	5,572	574
2015–2016	9,971	5,474	5,448	598
2017–2018	9,254	5,265	5,230	647
Total	39,156	21,646	**21,552**	**2,369**

Outcome. CVD = 1 if the participant reported ever being told by a doctor they had congestive heart failure (MCQ160B), coronary heart disease (MCQ160C), angina (MCQ160D), heart attack (MCQ160E) or stroke (MCQ160F); 0 if "no" to all five; excluded otherwise (refused/don't know/missing; n = 94). Prevalence: 11.0 % unweighted, 8.9 % survey-weighted (MEC weight / 4, per NCHS guidance for combining cycles).

Features (20). Age; sex; race/ethnicity (4 indicators, ref. non-Hispanic White); income-to-poverty ratio; education; systolic and diastolic BP (mean of up to 3 readings, zero diastolic treated as missing); pulse rate; BMI; waist circumference; total cholesterol; HDL cholesterol; HbA1c; serum creatinine; uric acid; smoking status (never/former/current); diagnosed diabetes. Medication variables were deliberately **excluded** as features because treatment is frequently a consequence of diagnosed CVD (label leakage). Maximum missingness: income-to-poverty ratio 9.8 %; all other features ≤ 6.3 %.

Split and preprocessing. Stratified 80/20 split (train n = 17,241 / 1,895 cases; test n = 4,311 / 474 cases). Median imputation and standardisation were fitted inside each pipeline on training folds only (no leakage).

Models and tuning (5-fold stratified CV on the training set, objective ROC-AUC):

Model	Search space	Selected
Logistic Regression (L2)	C in {1e-3 … 100}	C = 0.01
Random Forest (500 trees)	depth {8,12,None} × min leaf {5,20,50} × max_features {sqrt,0.5}	depth 12, leaf 20, sqrt
SVM (RBF, Platt probabilities)	C {0.1,1,10,100} × γ {scale,0.01,0.001,0.0003} × class_weight {None,balanced}	C = 10, γ = 0.001, balanced
XGBoost	40-iteration random search (7 hyperparameters)	502 trees, depth 5, lr 0.0113
MLP (PyTorch)	5 architectures × dropout {0.1,0.3,0.5} × weight decay {1e-4,1e-2} × lr {1e-3,3e-4} (60 configs)	64-32, dropout 0.5, wd 0.01, lr 1e-3

MLP: Linear–BatchNorm–ReLU–Dropout blocks, AdamW, BCE loss, batch 256, early stopping on a 15 % internal validation split (patience 15); final model 3,649 parameters, stopped at epoch 89.

Decision threshold. Chosen per model to maximise F1 on out-of-fold training predictions, then applied unchanged to the test set (the test set was never used for any choice).

Metrics. ROC-AUC (1,000-sample bootstrap 95 % CI), PR-AUC, precision, recall, specificity, F1, Brier score, ECE (10 quantile bins), calibration slope and intercept; DeLong test for paired AUC differences; survey-weighted test AUC; tuning/fit/inference time; SHAP time.

SHAP. One model-agnostic method for all five models (permutation SHAP on the predicted probability; 300 test rows, 50 training background rows, 164 evaluations/row) so attributions are directly comparable. Validated against exact interventional TreeSHAP for XGBoost: importance-ranking Spearman ρ = **0.989**.

Stability. (i) Cross-model: pairwise Spearman/Kendall correlation of mean |SHAP| vectors, top-5 Jaccard, Kendall's W across all five models. (ii) Within-model: 5 bootstrap resamples of the training set, refit, recompute SHAP, pairwise Spearman and top-5 Jaccard.

Ablation. Leave-one-feature-group-out (8 groups), retrained with tuned hyperparameters, ΔAUC averaged over 5 different random stratified splits. **Robustness:** same 5 splits for full-model AUC variability; **temporal validation:** train on 2011–2016 (n = 16,322), test on 2017–2018 (n = 5,230).

2. Results

Table 1 — Survey-weighted characteristics (`table1_characteristics.md`)

CVD cases were older (65.0 vs 46.1 y), more often had diagnosed diabetes (33.0 % vs 8.3 %) and were more often former smokers (38.0 % vs 23.1 %). Notably, CVD cases had **lower** total cholesterol (177.5 vs 193.0 mg/dL) and diastolic BP (67.8 vs 71.3 mmHg) — see §3.

Table 2 — Test-set performance (n = 4,311; 474 cases)

Model	CV AUC (SD)	Test AUC [95 % CI]	PR-AUC	Precision	Recall	F1	Brier	ECE	Cal. slope	DeLong p vs MLP
Logistic Regression	0.846 (0.009)	0.839 [0.821–0.855]	0.392	0.383	0.561	0.455	0.080	0.010	1.04	0.321
Random Forest	0.847 (0.003)	0.839 [0.821–0.856]	0.407	0.361	0.586	0.447	0.080	0.010	1.16	0.460
SVM (RBF)	0.849 (0.007)	0.839 [0.821–0.854]	0.396	0.362	0.584	0.447	0.080	0.007	0.97	0.274
XGBoost	0.850 (0.005)	**0.848 [0.829–0.863]**	**0.418**	0.373	0.572	0.451	**0.078**	0.009	1.04	0.064
MLP	0.849 (0.007)	0.842 [0.825–0.858]	0.398	0.390	0.523	0.447	0.080	0.018	0.91	—

- XGBoost had the highest AUC; it was significantly better than LR (p = 0.005), RF (p < 0.001) and SVM (p = 0.002), but **not** significantly better than the MLP (p = 0.064).
- The MLP was **not** significantly different from any classical model (all DeLong p ≥ 0.064).
- All models span only 0.009 AUC; F1 spans 0.447–0.455.
- All models were well calibrated (ECE ≤ 0.018). The MLP had the largest ECE and a slope < 1 (slightly over-confident); RF had a slope > 1 (slightly under-confident).
- Survey-weighted test AUC: 0.851–0.862, same ordering.

Repeated splits (5 seeds, `repeated_split_auc.md`): mean AUC LR 0.842 ± 0.002, RF 0.841 ± 0.004, SVM 0.843 ± 0.003, XGB 0.846 ± 0.004, MLP 0.844 ± 0.003 — differences are within one SD.

Temporal validation (train 2011–2016 → test 2017–2018, `temporal_validation.md`): AUC LR 0.838, RF 0.837, SVM 0.838, XGB 0.838, MLP 0.839; Brier 0.088 for all. No model degraded meaningfully over time and the ranking advantage of XGBoost disappeared.

Computational efficiency (`table_efficiency.md`)

Model	Tuning (s)	Final fit (s)	Inference (ms / 1k rows)	SHAP, 300 rows (s)
LR	5	0.05	1.1	10
RF	271	1.5	24.8	120
SVM	224	47.4	828	1,615
XGB	49	0.6	2.9	10
MLP	352	9.0	1.2	5

The MLP was the most expensive model to tune and ~170× slower to fit than LR for no significant accuracy gain. SVM was by far the slowest at inference and explanation. (Wall-clock times, 12-core CPU; tuning ran in parallel.)

Figure 2 / Table 3 — SHAP feature importance and stability

- **Age was the #1 feature for all five models.** The same five features — age, smoking, total cholesterol, diagnosed diabetes, income-to-poverty ratio — form the top 5 for LR, SVM and MLP; RF and XGB swap in

serum creatinine.

- Overall cross-model concordance: **Kendall's W = 0.746**.
- **Agreement clusters by model family, not by "classical vs. deep":** RF–XGB ρ = 0.941 (tree ensembles); SVM–MLP ρ = 0.886 and LR–MLP ρ = 0.743 (smooth decision functions); cross-family pairs ρ = 0.47–0.66. The MLP's attributions are closer to LR/SVM than the tree models are.
- **Within-model stability (bootstrap refits):** RF ρ = 0.973, XGB 0.921, LR 0.846, MLP 0.805, SVM 0.732. Tree models give the most stable full rankings, but LR/SVM/MLP keep a perfectly stable top 5 (Jaccard 1.00 for SVM and MLP, 0.87 for LR) whereas RF/XGB top-5 sets vary (0.77 / 0.73) because several mid-ranked features have near-equal importance.
- Tree models assign more importance to continuous lab/physiological measures (creatinine, HbA1c, systolic BP), while LR/SVM/MLP assign more to demographics (ethnicity indicators, smoking).

Figure 4 / Table 4 — Leave-one-group-out ablation (ΔAUC, mean ± SD over 5 splits)

- Removing **demographics** caused the largest drop for every model (−0.039 to −0.063), largest for LR (−0.063), i.e., LR relies most on age; nonlinear models partially recover the signal from correlated clinical variables.
- Lipids, glycaemic, socioeconomic, renal/metabolic and smoking groups each cost 0.002–0.007 AUC.
- **Anthropometric and blood-pressure groups contributed ≈ 0** ($|\Delta AUC| \leq 0.0013$) for every model.
- The ranking of groups was nearly identical across all five models — ablation agrees with SHAP at the group level.

Figures

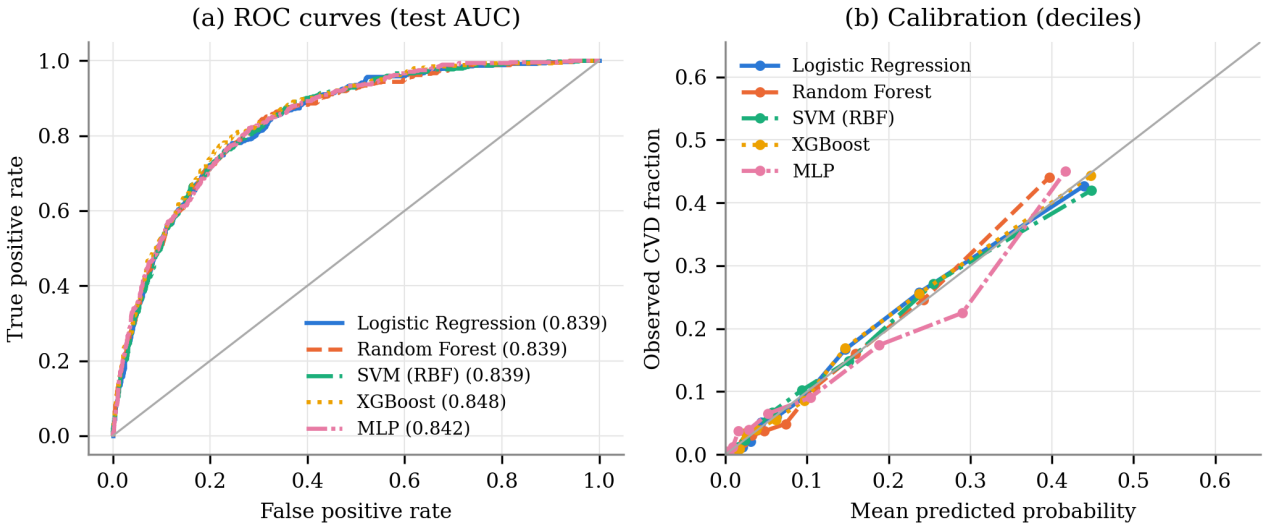


Figure 1. (a) ROC curves on the held-out test set (n = 4,311), AUC in the legend. (b) Calibration by decile of predicted risk.

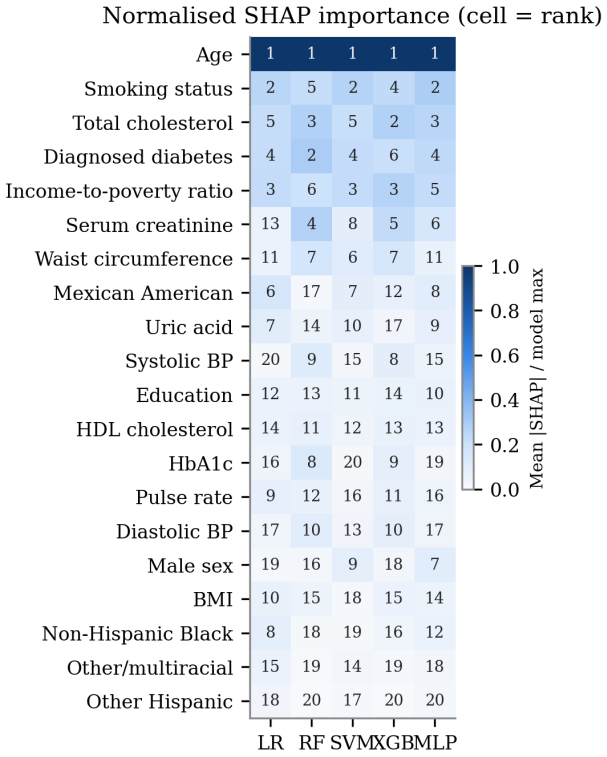


Figure 2. Mean |SHAP| per model, scaled to each model's top feature; numbers are within-model ranks.

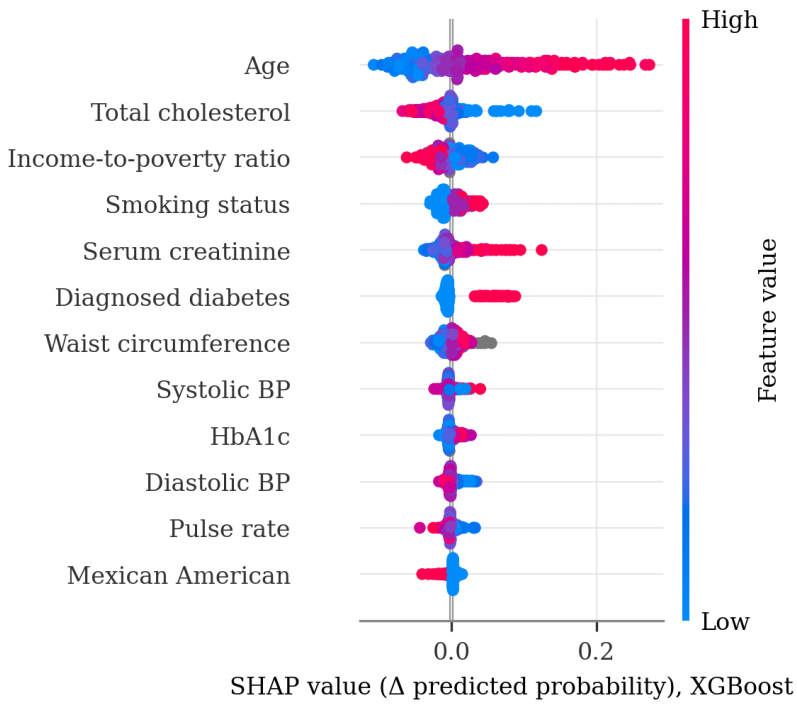


Figure 3. SHAP beeswarm for XGBoost (best model). Each dot is one test participant; grey dots are missing values.

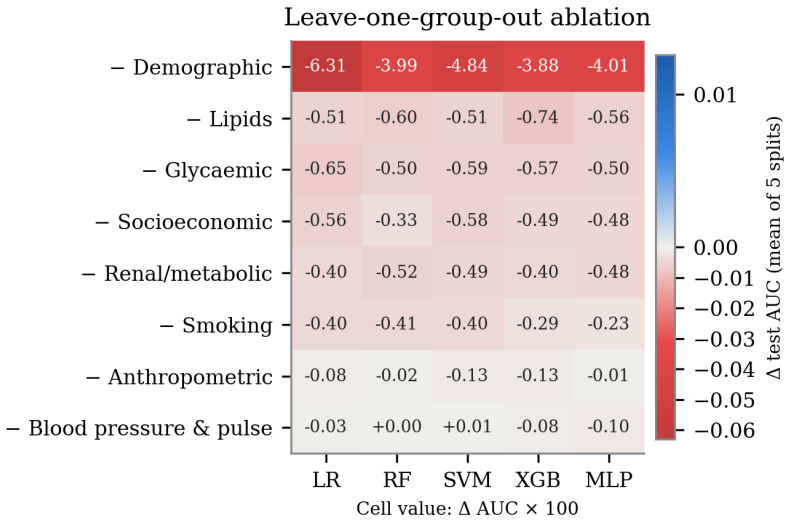


Figure 4. Change in test AUC (× 100) when each feature group is removed, mean of 5 random splits.

3. Discussion points (non-causal)

1. **Deep learning gave no measurable advantage.** The MLP was statistically indistinguishable from LR, RF and SVM and non-significantly below XGBoost, consistent with prior evidence that gradient-boosted trees match or exceed neural networks on tabular data. Under temporal validation all five converge to AUC ≈ 0.838.
2. **Simplicity is competitive.** LR reached within 0.009 AUC of the best model, was well calibrated, and was ~170× cheaper to train than the MLP.
3. **Attribution agreement depends on model family.** Similar AUC does not imply similar explanations; tree ensembles and smooth models distribute credit differently among correlated features. Reporting SHAP from a single model would give an incomplete picture.
4. **Reverse causation from treatment (important caveat).** SHAP shows *higher* total cholesterol associated with *lower* predicted CVD probability in all models. This is explained by treatment after diagnosis (`06_medication_check.py`): weighted statin use 58.0 % in CVD cases vs 14.2 % in non-cases; among statin users mean cholesterol was 166.4 mg/dL (CVD) vs 183.0 (no CVD). Similarly 62.8 % of cases vs 20.1 % of non-cases take BP medication, which likely explains why blood pressure added ≈ 0 AUC. Because the outcome is **prevalent, self-reported** CVD measured cross-sectionally, models learn markers of *having been diagnosed and treated*, not *risk of developing* CVD. Attributions describe predictive association only.
5. **Limitations.** Cross-sectional, self-reported outcome (recall bias, no incidence); age top-coded at 80; survey weights used for descriptive statistics and a weighted AUC sensitivity check, but not in model training; permutation SHAP is approximate (validated ρ = 0.989 vs exact TreeSHAP); for the MLP, the selected dropout (0.5) and weight decay (0.01) lie on the edge of the search grid, although the selected architecture (64-32) is interior and the CV difference between the top configurations is small.

4. Results-based abstract (replace the "proposed study" abstract)

Abstract— Cardiovascular disease (CVD) classification from routinely collected clinical data offers a setting to compare the accuracy and interpretability of machine-learning methods. Using four cycles (2011–2018) of the National Health and Nutrition Examination Survey (n = 21,552 adults; 2,369 with self-reported CVD), we compared Logistic Regression, Random Forest, Support Vector Machine and XGBoost with a multilayer perceptron (MLP) on 20 harmonized demographic, clinical and laboratory features. All models achieved similar discrimination on a held-out test set (ROC-AUC 0.839–0.848); XGBoost performed best (0.848, 95 % CI 0.829–0.863), and the MLP (0.842) did not differ significantly from any classical model (DeLong $p \geq 0.064$). All models were well calibrated (ECE ≤ 0.018), and under temporal validation on 2017–2018 data every model reached AUC ≈ 0.838. The MLP required the longest tuning time and about 170 times the training time of Logistic Regression. SHAP analysis identified age as the most influential feature for every model, and feature rankings were broadly concordant (Kendall's W = 0.75), but agreement clustered by model family: tree ensembles agreed closely with each other (ρ = 0.94), as did smooth-boundary models (SVM–MLP ρ = 0.89). Feature-group ablation confirmed the dominance of demographic variables. Inverse attributions for cholesterol were traced to post-diagnosis treatment, underscoring that attributions reflect predictive association, not causation. For structured clinical survey data, deep learning provided no measurable advantage over classical approaches.

5. Files

Script	Purpose
`01_build_dataset.py`	Merge & harmonize 4 cycles → `data/nhanes_cvd_2011_2018.csv`, `data/cohort_flow.csv`
`common.py`	Features, pipelines, PyTorch MLP, metrics, DeLong, bootstrap
`02_train_evaluate.py`	Tuning, OOF thresholds, test metrics, DeLong → `results/test_metrics.csv`, `models/`
`03_shap_stability.py`	SHAP for 5 models, TreeSHAP check, cross/within-model stability
`04_ablation_robustness.py`	Group ablation, repeated splits, temporal validation
`05_figures_tables.py`	Figures 1–4 and all tables
`06_medication_check.py`	Supplementary statin / BP-medication check

Figures: `fig1_roc_calibration` (ROC + calibration), `fig2_shap_importance_heatmap`, `fig3_shap_beeswarm_XGB`, `fig4_ablation`.

References for the paper (IEEE style)

Numbered in the order they would first be cited (Introduction → Related Work → Methodology → Discussion), as IEEE requires. Each entry was checked against Crossref or the publisher/PubMed Central page in October 2026. DOIs are given wherever one exists. Entries marked † are conference/JMLR papers that have no DOI; they were checked against the proceedings or journal page.

A 6-page IEEE paper usually has 20–30 references. The **core set** (*) is 25 entries and fits about 0.9 page. The rest are optional extras.

A. Introduction: CVD burden and clinical risk scores

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B. Related Work: machine learning for CVD

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NHANES-based studies (closest to this paper)

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Classical ML vs. deep learning on tabular / clinical data

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C. Methodology: data, models, tools

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Evaluation metrics and statistics

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SHAP and stability

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D. Discussion: reporting standards

* [43] G. S. Collins, K. G. M. Moons, P. Dhiman, *et al.*, "TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024, doi: 10.1136/bmj-2023-078378.

[44] E. W. Steyerberg and Y. Vergouwe, "Towards better clinical prediction models: Seven steps for development and an ABCD for validation," *Eur. Heart J.*, vol. 35, no. 29, pp. 1925–1931, 2014, doi: 10.1093/eurheartj/ehu207.

> Cite TRIPOD+AI [43], not the 2015 TRIPOD statement. TRIPOD+AI officially replaces it, and reviewers check this.

Related Work summary table (for Section II)

Your outline asks Related Work to compare 5–8 studies and say what they did not evaluate. This table does that.

Study	Data	Models	Explainability	What it did not do (gap this paper fills)
Weng *et al.* 2017 [6]	UK CPRD, 378k patients	LR, RF, GBM, NN vs ACC/AHA score	none	No SHAP; no calibration comparison; no attribution agreement across models
Ambale-Venkatesh *et al.* 2017 [8]	MESA, 6.8k	Random survival forest vs Cox	variable importance from one model	One model family; no deep learning comparison
Alaa *et al.* 2019 [7]	UK Biobank, 423k	AutoML ensemble vs Framingham	AutoML variable ranking	No head-to-head classical vs MLP; no explanation stability analysis
Dinh *et al.* 2019 [11]	NHANES 1999–2014	LR, SVM, RF, gradient boosting, ensemble	feature importance	No MLP; no calibration analysis; no cross-model attribution agreement
Krittanawong *et al.* 2020 [9]	meta-analysis	many	n/a	Pooled AUCs only; heterogeneous outcomes and no shared evaluation
Martin-Morales *et al.* 2023 [12]	NHANES 2003–2012, CVD death	LR, SVM, RF, XGB, LightGBM	SHAP	No deep learning; no attribution agreement across models; undersampled data
Ahiduzzaman & Hasan 2025 [13]	NHANES 2017–2023	LR, RF, SVM, XGB, LightGBM	SHAP + LIME	No MLP; no attribution stability across or within models
Christodoulou *et al.* 2019 [14]	systematic review, 71 studies	ML vs LR	n/a	Shows ML ≈ LR in clinical data; motivates this comparison

Gap statement you can use: previous NHANES CVD studies compared classical models or reported SHAP for one model. None (a) compared classical models with a tuned MLP under the same pipeline, (b) measured whether SHAP attributions agree across model families and stay stable across refits, or (c) checked calibration and temporal validity together.

Where to cite what

- **Intro, paragraph 1 (burden):** [1], [2]
- **Intro (risk scores, nonlinearity):** [3], [4], [5]
- **Intro (DL not guaranteed better on tabular data):** [14]–[16]
- **Intro (explainability, attribution ≠ causation):** [18], [39]
- **Methods, data:** [21], [22]
- **Methods, models:** LR + [31]; RF [23]; SVM [24], [25]; XGB [26]; MLP [27]–[29], [32]; tuning [30], [31]
- **Methods, metrics:** [33]–[38]
- **Methods, SHAP:** [39], [40]; Kendall's W [41]
- **Discussion (performance parity):** [14]–[17]
- **Discussion (family-dependent attributions):** [20], [42]
- **Discussion (calibration and reporting):** [38], [43], [44]